

Alessandro Giacchetto

Rita Levi Montalcini Researcher · University of Padua

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Employment

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|-------------------|---|
| Nov 2026–present | Rita Levi Montalcini Researcher , University of Padua |
| Oct 2023–Oct 2026 | ETH Fellow & Hermann-Weyl-Instructor , ETH Zürich
Mentor: Rahul Pandharipande |
| Oct 2021–Oct 2023 | Postdoctoral Fellow , IPhT
Mentor: Bertrand Eynard |

Education

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| 2021 | PhD in Mathematics , MPIM
<i>Geometric and topological recursion and invariants of the moduli space of curves</i>
Supervisor: Gaëtan Borot |
| 2017 | MSc in Mathematics , University of Trieste & SISSA
<i>The J-equation on Kähler manifolds and blowups</i>
Supervisor: Jacopo Stoppa |
| 2015 | BSc in Mathematics , University of Trieste
<i>Skein relations and polynomial invariants of knots and links</i>
Supervisor: Bruno Zimmermann |

Publications and preprints

Membranes and maps R. Pandharipande, Y. Schuler · Preprint · 2026  [arXiv](#)

A bijective topological recursion for maps G. Borot, L. Merkens · Preprint · 2026  [arXiv](#)

Refined lattice point counting on the moduli space of Klein surfaces N. K. Chidambaram, E. Garcia-Failde, K. Osuga · Preprint · 2026  [arXiv](#)

Worksheet duals to one-matrix models R. Gopakumar, E. A. Mazenc · Preprint · 2026
 [arXiv](#)  [Notebook](#)

Theta classes: generalized topological recursion, integrability and $\mathcal{W}$ -constraints V. Bouchard, N. K. Chidambaram, S. Shadrin · Preprint · 2025  [arXiv](#)

Les Houches lecture notes on moduli spaces of Riemann surfaces D. Lewański · Les Houches Summer School Lecture Notes - 2024-08: Quantum Geometry · *SciPost Phys. Lect. Notes* · 2025
 [arXiv](#)  [Open access](#)

The factorial growth of topological recursions G. Borot, B. Eynard · *Lett. Math. Phys.* 115.62 · 2025
 [arXiv](#)  [DOI](#)

Can transformers do enumerative geometry? B. Hashemi, R. G. Corominas · *13th Int. Conf. Learn. Represent.* · 2025  [arXiv](#)  [Open access](#)  [Code](#)

Symmetries of F -cohomological field theories and F -topological recursion G. Borot, G. Umer · *Commun. Math. Phys.* 406.248 · 2025  [arXiv](#)  [DOI](#)

Length spectrum of large genus random metric maps S. Barazer, M. Liu · *Forum Math. Sigma* 13 · 2025  [arXiv](#)  [DOI](#)

Resurgent large genus asymptotics of intersection numbers B. Eynard, E. Garcia-Failde, P. Gregori, D. Lewański · Preprint · 2023  [arXiv](#)

The spin Gromov–Witten/Hurwitz correspondence for $\mathbb{P}^1$ R. Kramer, D. Lewański, A. Sauvaget · *J. Eur. Math. Soc.* · 2025  [arXiv](#)  [DOI](#)

Relations on $\overline{\mathcal{M}}_{g,n}$ and the negative (r) -spin Witten conjecture N. K. Chidambaram, E. Garcia-Failde · *Invent. Math.* 241 · 2025  [arXiv](#)  [DOI](#)

Shifted Witten classes and topological recursion S. Charbonnier, N. K. Chidambaram, E. Garcia-Failde · *Trans. Amer. Math. Soc.* 377.2 · 2024  [arXiv](#)  [DOI](#)

An intersection-theoretic proof of the Harer–Zagier formula D. Lewański, P. Norbury · *Algebraic Geom.* 10.2 · 2023  [arXiv](#)  [DOI](#)

Around the combinatorial unit ball of measured foliations on bordered surfaces G. Borot, S. Charbonnier, V. Delecroix, C. Wheeler · *Int. Math. Res. Not.* 2023.17 · 2023  [arXiv](#)  [DOI](#)

A new spin on Hurwitz theory and ELSV via theta characteristics R. Kramer, D. Lewański · Accepted in *Selecta Math.* · 2025  [arXiv](#)

On the Kontsevich geometry of the combinatorial Teichmüller space J. E. Andersen, G. Borot, S. Charbonnier, D. Lewański, C. Wheeler · Preprint · 2020  [arXiv](#)

Masur–Veech volumes and intersection theory, the principal strata of quadratic differentials G. Borot, D. Lewański · Appendix in *Duke Math. J.* 172.9 · 2023  [arXiv](#)  [DOI](#)

Topological recursion for Masur–Veech volumes J. E. Andersen, G. Borot, S. Charbonnier, V. Delecroix, D. Lewański, C. Wheeler · *J. London Math. Soc.* 107.1 · 2023  [arXiv](#)  [DOI](#)

Teaching

[Toric Geometry](#), lecturer, ETHZ, Spring 2025

Moduli spaces of Riemann surfaces, PhD mini-course, Les Houches, Summer 2024

[Riemann Surfaces](#), lecturer, ETHZ, Spring 2024

Hurwitz numbers and Gromov–Witten theory, PhD mini-course, Hiddensee, Summer 2022

Topology, teaching assistant, UniTS, Spring 2017

Linear algebra, teaching assistant, UniTS, Fall 2016

Organization of events

Workshop [Refinement in Enumerative Geometry and Physics](#), Bangalore (22 Jun–3 Jul, 2026)

School [Quantum Geometry](#), Les Houches (29 Jul–23 Aug, 2024)

School [Topological recursion and integrability](#), Trieste (11–16 Sep, 2023)

Conference [Moduli spaces: theory and coding](#), Les Diablerets (27 Feb–3 Mar, 2023)

School + workshop [TR Salento 2022](#), Otranto (5–16 Sep, 2022)

Conference [Recent advances on moduli spaces of curves](#), Leysin (18–24 Mar, 2022)

Reading group [Donaldson–Thomas invariants](#), MPIM (Spring 2020)

Reading group [Integer points in polyhedra](#), MPIM (Fall 2019)

Reading group [Geometric recursion](#), MPIM (Fall 2018)

Honours

2025	<u>SwissMAP Junior Researcher Prize</u>
Oct 2023–Sep 2025	ETH Fellowship
2018	Marco Reni prize
Sep 2015–Sep 2017	SISSA MSc Fellowship
2012/13	Best freshman, University of Trieste
Sep 2012–Sep 2015	Luciano Fonda BSc Fellowship